

Many Producers Still Require One Actuator Writer

Project Hands-on Day 4: Make Actuation Safe

`runtime_poll_v1` $\longrightarrow$ `safety_v1`

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`runtime_poll_v1` must be green before safety work begins

Input `runtime_poll_v1` $\longrightarrow$ **Output** `safety_v1`

Run once from a clean tree

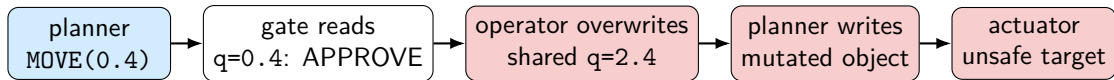
```
make verify-day3
```

Record the commit, compiler, seed, and the 50-client workload. On failure, restore the latest class-generated `runtime_poll_v1` checkpoint.

The inherited contract is ready when

- fast and safety paths ignore the slow client;
- four timestamps, queue age, and today's fault fixtures are present.

A command can pass the gate and still lose the write race



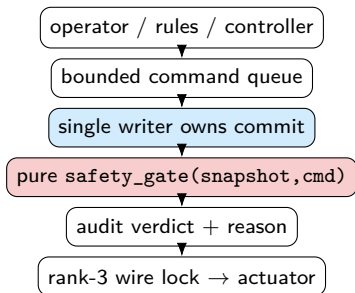
Reproduce one failure: run the supplied `check-write-race` fixture until the actuator trace contains a sequence with no matching approved snapshot.

Record this signature

- approved sequence $\neq$ written sequence;
- two writer IDs or interleaved bytes;
- missing verdict/reason in audit.

The bug is the mutable gap between decision and write—not the range check.

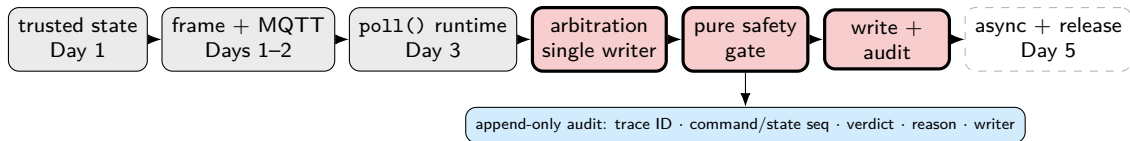
One immutable snapshot, one writer, and one verdict close the gap



Soda Code

```
fn commit_one(c: Cmd) {  
    let s = state_snapshot();  
    let (verdict, reason) = safety_gate(&s, &c);  
    let approved = verdict == Verdict::Approve;  
    audit_append(c.seq, s.seq, verdict, reason);  
    if !approved {  
        return;  
    }  
  
    let mut wire = WIRE.lock()  
        .expect("wire mutex poisoned");  
    wire.write_frame(&c);  
}
```

Today inserts one safe commit path and freezes every earlier contract



Change today

arbiter, safety_gate, actuator_writer,
audit

Frozen from Days 1-3

state/command schema, FrameV1,
topics/timestamps, buffer and queue policy

Five teams feed one common safety verdict

Team	Build	Deliver	Pass
G1 1/2	Connect state uncertainty/freshness; test stale and occluded poses.	Threshold evidence + verdict rows	Low-quality state never actuates.
G2 3/4	Replay corrupt/stale input through arbitration and safety.	Cross-layer fault log	Bad transport input stays rejected.
G3 5/6	Add single-writer queue, lock-wait trace, audit hook to poll().	Runtime+safety trace	Every actuation has one writer.
G4 7/8	Implement arbitration, lock order, pure gate, watchdog, audit.	safety_v1 fault table	Unsafe=0; valid NACK=0.
G5 9/10	Integrate arbitration, safety, audit into the gatekeeper.	Five-gate end-to-end trace	Unsafe/corrupt/stale all reject.

Race, reject, and record every decision

Soda Code

```

1 fn safety_gate(s: &State, c: &Cmd)
2   -> Result<(), Reason> {
3   if !state_fresh(s, MAX_STATE_AGE_NS) {
4     return Err(Reason::StaleState);
5   }
6   // TODO 1: reject non-finite, range/rate,
7   // joint, uncertainty; return Err(reason).
8   Ok(())
9 }
10 fn actuator_submit(c: Cmd) {
11   // TODO 2: move c into the bounded writer queue.
12   // Only actuator_pump may gate, audit, write.
13 }

```

Minute 11: green / blocked / using class checkpoint. **Frozen:** thresholds, enum values, state fields, and transport contracts.

Fixed fixtures

F1 check–write race
 F2 reversed lock order
 F3 NaN command
 F4 joint/rate violation
 F5 stale state

Roles: A–gate; B–writer; C–interface and raw evidence.

Safety passes only when every rejection has evidence

Run the canonical gate

```
make verify-day4
```

Save these raw files:

- raw/fault_verdicts.csv
- raw/lock_trace.jsonl
- raw/audit.log
- manifest.json

Frozen check	Required result
Unsafe actuator writes	0
Spurious NACKs on valid traffic	0
Rejects with stable verdict + reason	100%
Rejects represented in append-only audit	100%
Race and deadlock after repair	terminate; no bypass
Actuation records with one writer ID	100%

Then add 2–3 claim/method bullets and one figure/table caption per chapter.

Tomorrow, emergency work must preempt ordinary work

Input `runtime_poll_v1` $\longrightarrow$ **Output** `safety_v1`

Tag only after the gate is green

```
git tag safety_v1
```

Record the artifact owner, commit, manifest path, and one known issue.

Freeze for Day 5

- safety thresholds and `MAX_STATE_AGE_NS`;
- verdict/reason enum values;
- audit fields and single-writer invariant;
- class `safety_v1` checkpoint.

Thanks for listening

Presented by Suyu Jiang

Template: Azusa Template